

image

7x7 Conv, 64, /2

max pool, /2

3x3 Conv, 64

3x3 Conv, 64

3x3 Conv, 64

3x3 Conv, 64

3x3 Conv, 128, /2

3x3 Conv, 128

3x3 Conv, 128

3x3 Conv, 256, /2

3x3 Conv, 256

3x3 Conv, 256

3x3 Conv, 512, /2

3x3 Conv, 512

3x3 Conv, 512

Average pool

FC, 10

Architecture Diagram of ResNet 18.

5/11/25 Exp: 13

Understanding a Pre trained model.

Aim: The Primary aim of this experiment is to load inspect the complete architecture of a Pre-trained ResNet-18 model using PyTorch.

Objectives:

- To import the libraries and Understand its Use
- To fetch the ResNet-18 model architecture.
- Print the summary and Print the architecture.

Pseudo Code.

- Import the necessary libraries like torch, torchvision and Summary from torch Summary
- Load the ResNet-18 model from the model library
- Include the weight Pre-trained on ImageNet
- Set newly loaded mode to evaluation mode.
- After evaluation Print the model architecture and the model summary.

Output:

Total Parameters: 11, 689, 512

Trainable Parameters: 11, 689, 512

Non Trainable Params: 0

Layers Name

Conv1: Conv2d

bn1: Batch Norm2d

relu: ReLU

maxpool: Max Pool2d

layer 1: Sequential

layer 2: Sequential

layer 3: Sequential

layer 4: Sequential

avg pool: Adaptive Avg Pool 2d

fc: Linear

2) The Summary is calculated based on sample input size of 3 channels, 22 is pixels height and 224 pixels width.

Result!

Successfully understood the structure of a Pretrained model.